

Task 2: Automate Local Gitea Project Setup

Objective

The objective of Task 2 was to create a shell script that automates the process of building and running the Gitea project locally without manually executing each command.

The automation was completed without using Docker and without hard-coding any user-specific paths.

About the Automation

A shell script named `setup.sh` was created in the root directory of the Gitea project.

The script automates the following tasks:

- Checks whether the required tools are available.
 - Displays the installed dependency versions.
 - Verifies that the script is running from the correct Gitea project directory.
 - Builds Gitea from source.
 - Verifies that the Gitea binary was created successfully.
 - Checks whether port 3000 is available.
 - Starts the Gitea web server.
 - Displays the local application URL.
 - Provides error handling and clear progress messages.
-

Environment Used

The project was completed on:

- Operating System: Windows
- Terminal Environment: MSYS2 UCRT64
- Shell: Bash

MSYS2 UCRT64 was used as the development environment because it provides a Unix-like terminal environment on Windows and allows Bash commands and development tools to be used for building and running the project.

No Docker was used during the setup.

Required Tools

The script checks for the following required tools:

- Git
- Go
- Node.js
- pnpm
- Make
- curl

The script uses the `command -v` command to verify whether each required tool is available in the system PATH.

If a required command is missing, the script displays an error message and stops execution.

Dependency Version Check

After verifying the required tools, the script displays their installed versions.

The following commands are used:

```
git --version
go version
node --version
pnpm --version
make --version
curl --version
```

Displaying the versions helps verify that the required development environment is available before starting the build process.

Project Directory Verification

The script verifies that it is being executed from the Gitea project root directory.

The following important project files are checked:

```
go.mod
Makefile
main.go
```

If these files are not found, the script stops and displays an error message.

This prevents the script from being accidentally executed from an incorrect directory.

No user-specific or hard-coded paths are used.

Script Implementation

The automation script was created as:

```
setup.sh
```

The script was made executable using:

```
chmod +x setup.sh
```

The script can then be executed using:

```
./setup.sh
```

Build Process

The script builds Gitea from source using:

```
make build
```

The build command automatically compiles the Gitea application.

The script checks whether the build command completes successfully.

If the build fails, an error message is displayed and the script stops.

When the build completes successfully, the script displays a success message.

Gitea Binary Verification

After the build process, the script verifies that the Gitea executable was created successfully.

Because the project was built on Windows, the generated executable is:

```
gitea.exe
```

The script also supports the Linux/macOS binary name:

```
gitea
```

If neither binary is found, the script displays an error and stops.

Port 3000 Check

Before starting the Gitea web server, the script checks whether port 3000 is already in use.

The port check uses netstat.

If port 3000 is already being used by another application, the script displays an error message and stops.

This prevents Gitea from failing to start because another process is already using the required port.

If the port is available, the script continues to start Gitea.

Starting the Gitea Web Server

After successfully completing all checks and the build process, the script starts the Gitea web server.

On the Windows environment, the script runs:

```
./gitea.exe web
```

The application becomes available at:

`http://localhost:3000`

The application was successfully opened and verified in a web browser.

The server can be stopped by pressing:

`Ctrl + C`

in the terminal.

Automation Flow

The complete workflow of the script is:

Check Required Tools



Display Dependency Versions



Verify Gitea Project Directory



Build Gitea from Source

↓
Verify Gitea Binary
↓
Check Port 3000
↓
Start Gitea Web Server
↓
Open `http://localhost:3000`

Error Handling

The script includes error handling throughout the automation process.

The following situations are handled:

- A required command is not available.
- The script is executed from the wrong directory.
- The Gitea build fails.
- The Gitea binary is not created.
- Port 3000 is already in use.

The script uses:

```
set -e
```

to stop execution when an unexpected command error occurs.

Clear status messages are also displayed during each stage of execution.

Testing

The `setup.sh` script was tested from the root directory of the Gitea project.

The script successfully:

1. Checked the required tools.
2. Displayed the installed dependency versions.
3. Verified the correct Gitea project directory.
4. Built Gitea from source.
5. Verified the creation of `gitea.exe`.
6. Checked that port 3000 was available.
7. Started the Gitea web server.
8. Displayed the local URL.
9. Successfully opened the Gitea application in the browser.

The application was verified at:

`http://localhost:3000`

Issues Faced

During the Task 2 setup and automation process, some environment-related issues were encountered.

MSYS2 UCRT64 Warnings

Repeated warnings similar to the following appeared while running some commands:

```
shared_info::initialize: size of shared memory region changed
```

These warnings were related to the MSYS2 environment and did not prevent Git, the build process, the automation script, or the Gitea application from working successfully.

Port 3000 Already in Use During Testing

While testing the automation process, it was necessary to stop an already running Gitea server before running the script again.

The existing server was stopped using:

```
Ctrl + C
```

The port was then checked using:

```
netstat -ano | grep ":3000"
```

Once no process was listening on port 3000, the automation script was successfully executed.

Git and GitHub Workflow

A separate branch was created for Task 2:

```
gitea-task-2
```

The `setup.sh` automation script was added, committed, and pushed to the GitHub repository.

The branch is configured to track:

```
origin/gitea-task-2
```

The changes were pushed successfully to the remote GitHub repository.

The Task 2 Pull Request will be raised within the user's own GitHub repository using the following branch comparison:

```
gitea-task-2 → main
```

No Pull Request will be raised to the official Gitea repository.

Files Added

The main automation file created for Task 2 is:

```
setup.sh
```

This documentation file was also created to document the Task 2 work:

```
task-2.md
```

What I Learned

Through this task, I gained practical experience with:

- Bash shell scripting.
 - Automating repetitive development tasks.
 - Checking whether commands are available using command `-v`.
 - Displaying and verifying dependency versions.
 - Validating the correct project directory.
 - Building a Go-based project from source.
 - Verifying generated binaries.
 - Checking whether a network port is available.
 - Starting a web application from a shell script.
 - Using error handling in Bash scripts.
 - Working with MSYS2 UCRT64 on Windows.
 - Managing Git branches and pushing changes to GitHub.
 - Following a feature branch and Pull Request workflow.
-

Status

The Task 2 automation script was successfully created and tested.

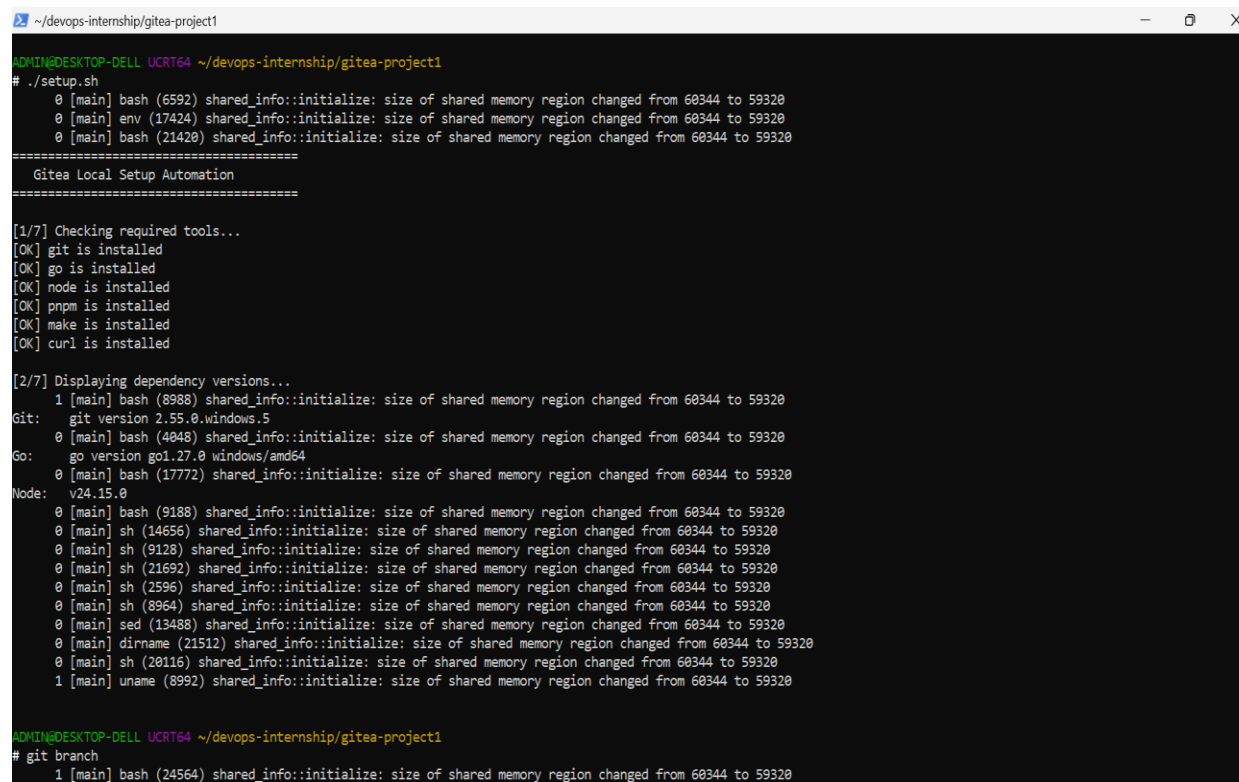
The script successfully automates the local Gitea setup process, including dependency checks, version verification, directory validation, project building, binary verification, port checking, and starting the Gitea web server.

The application was successfully verified in the browser at:

<http://localhost:3000>

Task 2 implementation and documentation have been completed successfully.

Screenshots:



```
~/devops-internship/gitea-project1
ADMIN@DESKTOP-DELL UCR764 ~/devops-internship/gitea-project1
# ./setup.sh
0 [main] bash (6592) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] env (17424) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] bash (21420) shared_info::initialize: size of shared memory region changed from 60344 to 59320
=====
Gitea Local Setup Automation
=====

[1/7] Checking required tools...
[OK] git is installed
[OK] go is installed
[OK] node is installed
[OK] npm is installed
[OK] make is installed
[OK] curl is installed

[2/7] Displaying dependency versions...
1 [main] bash (8988) shared_info::initialize: size of shared memory region changed from 60344 to 59320
Git: git version 2.55.0.windows.5
0 [main] bash (4048) shared_info::initialize: size of shared memory region changed from 60344 to 59320
Go: go version go1.27.0 windows/amd64
0 [main] bash (17772) shared_info::initialize: size of shared memory region changed from 60344 to 59320
Node: v24.15.0
0 [main] bash (9188) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (14656) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (9128) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (21692) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (2596) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (8964) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sed (13488) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] dirname (21512) shared_info::initialize: size of shared memory region changed from 60344 to 59320
0 [main] sh (20116) shared_info::initialize: size of shared memory region changed from 60344 to 59320
1 [main] uname (8992) shared_info::initialize: size of shared memory region changed from 60344 to 59320

ADMIN@DESKTOP-DELL UCR764 ~/devops-internship/gitea-project1
# git branch
1 [main] bash (24564) shared_info::initialize: size of shared memory region changed from 60344 to 59320
```


Internship Guidance Plan

Bhavish - Dashboard - Gitea: G

Ask Gemini

← → ↺

http://localhost:3000

☆ 📄 \$ Action required ⋮

Issues Pull Requests Milestones Explore

🔔 + 🌐

Bhavish

Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug

Mon

Wed

Fri

Sun

0 contributions in the last 12 months

Less More

Repository

Organization

Repositories

+



There are no repositories yet.

No Activity

You are currently not following any repositories or users, so there is no content to display. You can explore repositories or users of interest from the links below.

[Explore repositories](#) · [Explore users](#)

Powered by Gitea Version: 1.28.0+dev-408-g9eb4a9afad Pace: 10ms Template: 6ms

🌙 Auto

🌐 English

[Licenses](#)

[API](#)

```
~/devops-internship/gitea-project1
2026/08/28 17:11:09 cmd/web.go:264:runWeb() [I] Starting Gitea on PID: 3936
2026/08/28 17:11:09 cmd/web.go:115:showWebStartupMessage() [I] Gitea version: 1.28.0+dev-408-g9eb4a9afad built with go1.27.0
2026/08/28 17:11:09 cmd/web.go:116:showWebStartupMessage() [I] * RunMode: prod
2026/08/28 17:11:09 cmd/web.go:117:showWebStartupMessage() [I] * AppPath: C:\msys64\home\ADMIN\devops-internship\gitea-project1\gitea.exe
2026/08/28 17:11:09 cmd/web.go:118:showWebStartupMessage() [I] * WorkPath: C:\msys64\home\ADMIN\devops-internship\gitea-project1
2026/08/28 17:11:09 cmd/web.go:119:showWebStartupMessage() [I] * CustomPath: C:\msys64\home\ADMIN\devops-internship\gitea-project1\custom
2026/08/28 17:11:09 cmd/web.go:120:showWebStartupMessage() [I] * ConfigFile: C:\msys64\home\ADMIN\devops-internship\gitea-project1\custom\conf\app.ini
2026/08/28 17:11:09 cmd/web.go:121:showWebStartupMessage() [I] Prepare to run web server
2026/08/28 17:11:10 routers/init.go:117:InitWebInstalled() [I] Git version: 2.55.0 (home: C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\home)
2026/08/28 17:11:10 modules/storage/storage.go:227:initAttachments() [I] Initialising Attachment storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\attachments
2026/08/28 17:11:10 modules/storage/local.go:217:initAvatars() [I] Initialising Avatar storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\avatars
2026/08/28 17:11:10 modules/storage/storage.go:243:initRepoAvatars() [I] Initialising Repository Avatar storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\repo-avatars
2026/08/28 17:11:10 modules/storage/storage.go:237:initLFS() [I] Initialising LFS storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\lfs
2026/08/28 17:11:10 modules/storage/storage.go:249:initRepoArchives() [I] Initialising Repository Archive storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\repo-archive
2026/08/28 17:11:10 modules/storage/storage.go:259:initPackages() [I] Initialising Packages storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\packages
2026/08/28 17:11:10 modules/storage/storage.go:270:initActions() [I] Initialising Actions storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\actions_log
2026/08/28 17:11:10 modules/storage/storage.go:274:initActions() [I] Initialising ActionsArtifacts storage with type: local
2026/08/28 17:11:10 modules/storage/local.go:48:NewLocalStorage() [I] Creating new Local Storage at C:\msys64\home\ADMIN\devops-internship\gitea-project1\data\actions_artif
acts
2026/08/28 17:11:10 routers/common/db.go:22:InitDBEngine() [I] Beginning ORM engine initialization.
2026/08/28 17:11:10 routers/common/db.go:29:InitDBEngine() [I] ORM engine initialization attempt #1/10...
2026/08/28 17:11:10 cmd/web.go:205:serveInstalled() [I] PING DATABASE sqlite3
2026/08/28 17:11:11 routers/init.go:138:InitWebInstalled() [I] ORM engine initialization successful!
2026/08/28 17:11:11 .../indexer/issues/indexer.go:78:InitIssueIndexer.1() [I] PID 3936: Initializing Issue Indexer: bleve
2026/08/28 17:11:11 .../indexer/stats/indexer.go:41:populateRepoIndexer() [I] Populating the repo stats indexer with existing repositories
2026/08/28 17:11:11 .../indexer/issues/indexer.go:155:InitIssueIndexer.2() [I] Issue Indexer Initialization took 14.1653ms
2026/08/28 17:11:12 cmd/web.go:330:listen() [I] Listen: http://0.0.0.0:3000
2026/08/28 17:11:12 cmd/web.go:334:listen() [I] AppURL (ROOT_URL): http://localhost:3000/
2026/08/28 17:11:12 cmd/web.go:337:listen() [I] LFS server enabled
2026/08/28 17:11:12 modules/graceful/server.go:53:NewServer() [I] Starting new Web server: tcp:0.0.0.0:3000 on PID: 3936
```